
Objective

Passionate about Artificial Intelligence and Machine Learning, I aim to design, apply and develop new loss functions, activation functions, computational techniques to solve real-world problems, particularly in AI-driven applications and data science for bioinformatics using Federated Learning, provides the physical insight of the natural science. This passion drives me toward Computer Science and its applications in Biological Sciences.

Education

- 2022-Present** Doctor of Philosophy (PhD) under Visvesvaraya PhD Scheme, Department of Computer Science, Aligarh Muslim University, Aligarh India.
CGPA: 9.00/10
- 2018-21** Master in Computer Science and Application, Department of Computer Science, Aligarh Muslim University, Aligarh India.
CGPA: 8.46/10
- 2014-18** Bachelor of Science (Honours), Department of Computer Science, Aligarh Muslim University, Aligarh India.
CGPA: 6.45/10

Publications:

1. K. Ali and A. Zafar, "Comparative study of Federated Learning and Machine Learning for Drug Discovery," in *2024 IEEE International Conference on Computer Vision and Machine Intelligence (CVMI)*, IEEE, Oct. 2024, pp. 1–6. doi: 10.1109/CVMI61877.2024.10781631.
2. K. Ali and A. Zafar, "Federated Learning Applications for Infectious Diseases: A Systematic Review," in *2nd International Conference on Communication, Computer Sciences and Engineering (IC3SE-2025)*, IEEE, March 2025. (*Waiting to be available online*).
3. Jamaluddin, K. Ali and S. K. Sharma, "JK Algorithm: A Novel Approach for Single Nucleotide Polymorphism detection of breast cancer causing genes," *2025 International Conference on Intelligent and Secure Engineering Solutions (CISES)*, Greater Noida Gautam Budh Nagar, India, 2025, pp. 136-142, doi: 10.1109/CISES66934.2025.11265687.
4. K. Ali, M. Adnan, M. Shamim, and A. Narendra, "Federated Learning: A Novel Framework for Parkinson's Disease Detection," in *Proc. 2026 2nd Int. Conf. Emerging Computational Intelligence (ICECI)*, 2026. (*Waiting to be available online*).
5. M. Adnan, K. Ali, M. Shamim, and N. Sarwar, "Empirical Evaluation of Self and Cross-Model Verification for LLM Hallucination Detection in Cybersecurity," in *Proc. 2026 2nd Int. Conf. Emerging Computational Intelligence (ICECI)*, 2026. (*Waiting to be available online*).
6. M. Shamim, K. Ali, M. Adnan, and A. Y. Hashmi, "An Experimental Study of Soft Computing Techniques for Physiological Pattern Detection," in *Proc. 2026 2nd Int. Conf. Emerging Computational Intelligence (ICECI)*, 2026. (*Waiting to be available online*).

Projects:

Master's Project: Aligarh Muslim University, Aligarh

1. Project's Topic: "Next Vehicle Detection in fog using thermal radiation by image processing"

Description: Road accidents in fog, especially in winter, claim numerous lives. Our thermal radiation-based image processing system detects nearby vehicles using infrared emissions, alerting drivers and preventing collisions effectively.

2. Project's Topic: "CCN(CovidCasesNotification)"

Description: Developed CCN, a COVID case notification system providing real-time updates on infection trends across counties, states, and districts, improving pandemic response strategies.

3. Project's Topic: RePay (android Application)

Description: RePay is an Android application designed to assist small shopkeepers and Kirana owners in managing credit and debit transactions through a digital ledger, with downloadable PDF transaction reports.

Project's Topic: Car Store Management System

Description: The Car Store Management System (CSMS) is designed to ensure efficient data storage, manipulation, and organization of car store products, minimizing time and resources for quick data retrieval.

Skills

Experimental Skills: Bioinformatics Analysis, Machine Learning & Deep Learning Model Development, Federated Learning Applications

Programming: Python (Advance), R, C/C++, Java (Advance & GUI), SQL

Scripting language: HTML, CSS, PHP

Software: Familiar with MatLab

Training/Workshops/Conferences/ Schools attended:

- Participated as Member of Organizing team in **International Workshop on Bridging Languages with Generative Models: Advances in Speech Technology**, at the Indian Institute of Technology (IIT) Indore from Dec 15-16, 2025.
- IEEE CIS Summer School on **Emerging Trends in Computational Intelligence, Deep Learning, and Large Language Models** at the Indian Institute of Technology (IIT) Indore from July 23 to 25, 2025.
- Paper presented in IEEE International Conference of Communication, Computer Sciences and Engineering (EC3SE), 19th-21st March 2025, at Amity University, Greater Noida Campus, India.
- Paper presented in 3rd IEEE International Conference on Computer Vision & Machine Intelligence (CVMI) 2024, at IIIT Allahabad, India
- Attended **ACM India PhD Clinic, ARCS 2025**, at PSG Collage of Technology, Coimbatore
- Workshop on the Review of Research Work of 119 Full-time PhD candidates under Phase-II of the Visvesvaraya PhD Scheme, scheduled during September 25-27, 2024, at IIT Mandi
- A Workshop on "Deep Learning and Generative AI" organized by the Interdisciplinary Center for Artificial Intelligence, Aligarh Muslim University, Aligarh.
- An online course on "Introduction to AI/ML Techniques in Drug Discovery" from Udemy

- An online course on “Advanced Bioinformatics Data Analysis: Python, R and Linus” on Udemy
- An IEEE conference on CVMI organized by IIIT Allahabad, Prayagraj UP India.
- A webinar on **AI** organized by APPWARS Technologies held on 25 June 2020.
- A webinar on **AWS** Organized by APPWARS Technologies held on 14 June 2020.
- Four weeks online training program on “Communication Skills Development Program” organized by REsolv Corporation, USA in Collaboration with the Department of Computer Science, Aligarh Muslim University, Aligarh UP INDIA.

Academic Achievements

- Awarded Travel Grant to attend **13th ACM iKKD CODS, 2025**, at IISER, Pune
- **Anveshan Setu Fellowship** provided by **ACM India**
- Awarded Travel Grant to attend **ACM India PhD Clinic, ARCS 2025**, at Coimbatore
- **Visvesvaraya Fellow**
- Qualified: **JUNIOR RESEARCH FELLOWSHIP (JRF 2021)**
- Qualified: **NATIONAL ELIGIBILITY TEST (NET) at the National level.**
- Qualified: **GRADUATE APTITUDE TEST IN ENGINEERING(GATE2021).**
- **Athletic**
- **Hostel Management**

Co-curricular and other activities

- Participated in Athletic Activities
- Hostel Manager: I have selected as Hostel Manager on the basis of academic qualification and an interview in 2019-20, I worked as Hostel Manager.

References:

Dr. Aasim Zafar
Professor, Department of Computer Science,
Aligarh Muslim University, Aligarh India
Email: azafar.cs@amu.ac.in

Dr. Mohammad Sajid
Asst. Professor Department of Computer Science,
Aligarh Muslim University, Aligarh India
Email: msajid.cs@amu.ac.in